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STAT 310-410: Categorical Data Analysis (Archived)

All categories



Ch. 1 HW Problems 15 and 16a (STA... #17

Hi Dr. Whalen and all, I'm having a hard tim... ✓

👍 0 ⌚ 2 years ago

▼ Jan 24th - Jan 30th

Ch 1 HW Problem 12d #15

I'm getting a CI of (0, .196) for 12d, but R a...

0 2 years ago

Note about Case Control Studies #14
Where its stated "If fact, if $P(Y = \text{yes} | x)$..."
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Readings #13
I think we're ahead of the readings in lectu...
👍 0 ⌚ 2 years ago

Homework turn in (per 1/25 office ho... #12
I was just wondering if I could turn Homew... 
👍 0 ⌚ 2 years ago

2.4.6 Question #11
I hadn't heard of chi-squared statistics bei...
👍 0 ⌚ 2 years ago ✓

HW 1 Question 8b

Are we interpreting this as asking for a dir...

✓ #10

1 online now

★ An instructor (Mena Whalen) endorsed this answer

Charles Hwang 85 🐥

For 15a, since $\hat{\pi} = \frac{Y}{n}$, we can substitute that as $\sigma(\hat{\pi}) = \sigma(\frac{Y}{n})$. Per the mathematical properties (from "STAT 404: Introduction to Probability & Statistics" or equivalent), we have that $\sigma(\frac{Y}{n}) = \frac{\sigma(Y)}{|n|}$, which is $\frac{\sqrt{n\pi(1-\pi)}}{n}$ (solve from there).

For 15b, I think it's because there's "more room" for a wider CI at 0.50 than at 0 or 1 ? I referenced the sketch I made in Problem 1.4b.

For 16a, there is more information in Section 1.3.1, but it's asking for the natural log of the probability function for the binomial distribution. I'm not sure if I can share the answer here, but the first line is $\ell(\pi) = \binom{n}{k} \pi^k (1 - \pi)^{n-k}$ and it's asking for $\ln(\ell(\pi))$.

2 replies  Reply

 **Mena Whalen** ★

This is great.

For 15b. thinking about to 1.4b is great, just think about where the variance $\sigma(\hat{\pi})$ takes its max/min value at. Lets think about when $\pi = 1$, so every observation must be a success, and the probability that the sample proportion equals the true proportion is 1.

↩ Reply

 **Linda Galib** 11 

Hi Charles and Dr. Whalen, thank you so much for this! I really appreciate you taking the time to write this all out. Thanks again and see you in class today.